

Spring 26, Ken Short  
revised May 4, 2026 7:28 pm

## **Laboratory 12: Signature Analyzer III - Additional Features for Signature Analyzer**

To be performed the week starting May 3rd.

### **Prerequisite Reading**

1. Chapter 10 of the textbook.
2. HP 5004A Signature Analyzer Operating and Service Manual: Sections 1-13 through 1-17, 3-3 through 3-10,

### **Purpose**

The overall purpose of this laboratory is to add features to the system of Laboratory 11 Task 2 to bring its operation closer to that of an actual HP5004A Signature Analyzer.

### **Tasks**

*Task 1: Gate Generation from Start and Stop Pulses*

*Task 2: Prescalar for Multiplexed Digit Driver*

*Task 3: Confirm Funny Hex Fonts Implemented Correspond Exactly to HP5004A Fonts*

*Task 4: Structural Signature Analyzer with Added Features*

### **Design Tasks**

#### ***Design Task 1: Gate Generation from Start and Stop Pulses***

The **linear feedback shift register (LFSR)** that implemented the CRC signature in Laboratory 11 had an external `gate` input. The LFSR only shifted when while the `gate` input was asserted.

In contrast, the HP5004A does not have an external `gate` input. Its probe provides inputs `start`, `stop`, and `clock`. Its front panel also provides outputs from maintained edge select configuration switches that indicate whether the signature analyzer uses positive edges or negative edges of the probe outputs to define the occurrence of the events associated with each `start`, `stop`, and `clock` input. A 1 from an edge select switch selects positive edge. A 0 from an edge select switch selects negative edge. Write a design description for an entity named `gate_gen` that has the `clock`, `start`, and `stop` signals from the probe and `clocke`, `starte`, and `stope` signals from the front panel configuration switches. `gate_gen` generates the `gate` output used in Laboratory 11.

Entity `gate_gen` entity must be implemented as one or more FSMs. All FSMs used must be written as three-process FSMs. Use synchronous resets on all FSMs. Draw a state diagram for each FSM entity used.

Create a separate project and write a complete program to implement the `gate_gen` entity. Write a testbench for `gate_gen`.

**Submit your properly commented design description and testbench source files as part of your prelab.**

### ***Design Task 2: Prescalar for Multiplexed Digit Driver***

In order to improve the maximum speed of the clock input from the **device under test (DUT)** that our Signature Analyzer can handle, we need to increase our Signature Analyzers system clock speed. However, we don't want to run the multiplexed display at a high clock rate. Thus, we need to separate our Signature Analyzer's system clock frequency from the frequency at which we clock the digit driver.

If you are using the block diagram of Laboratory 10 to implement digit driving, you need to add an enable input to the `digit_counter`. That way the `digit_counter` only counts on its system clock's positive edge, if it is enabled to count.

An entity must be added to generate the enable pulse from the your signature analyzer's system clock. In Exam 2 Problem 4, entity `prescalar` was described that does just that. Write a design description that implements `prescalar`.

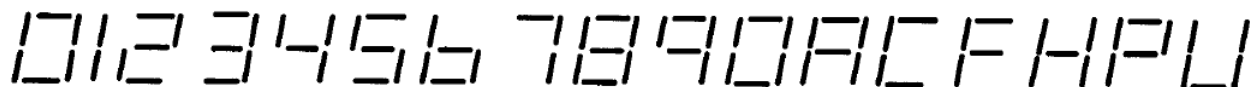
Also write a design description that modifies `digit_counter` so that it has an enable input.

Write a complete program for a structural style system that combines your `prescalar` and `digit_counter` to take a 4 Mhz system clock and generate the enable pulse to `digit_counter` at a 400 Hz rate. Write a non-self-checking testbench for this system.

**Submit your properly commented design description and testbench source files as part of your prelab.**

### ***Design Task 3: Confirm Funny Hex Fonts Implemented Correspond Exactly to HP5004A Fonts***

The fonts for funny hexadecimal in general have options for some characters. For example, a 6 could be represented with or without segment a turned ON. A 1 could be represented with either segments b and c ON or segments e and f ON. We would like our representation to be exactly the same as HP's. The following figure from the HP 5004A Signature Analyzer Operating and Service Manual shows exactly the fonts for the HP 5004A.



For a 1, use segments b and c. Note that in the HP manual's figure 0 appears twice. The second 0 should not be there.

Confirm that your `funnyhex_seven` module from Laboratory 10 produces exactly these fonts. If not, modify the code for that module so that it does.

Write a design description and non-self-checking testbench for this `funnyhex_seven`. Use your testbench to verify your design.

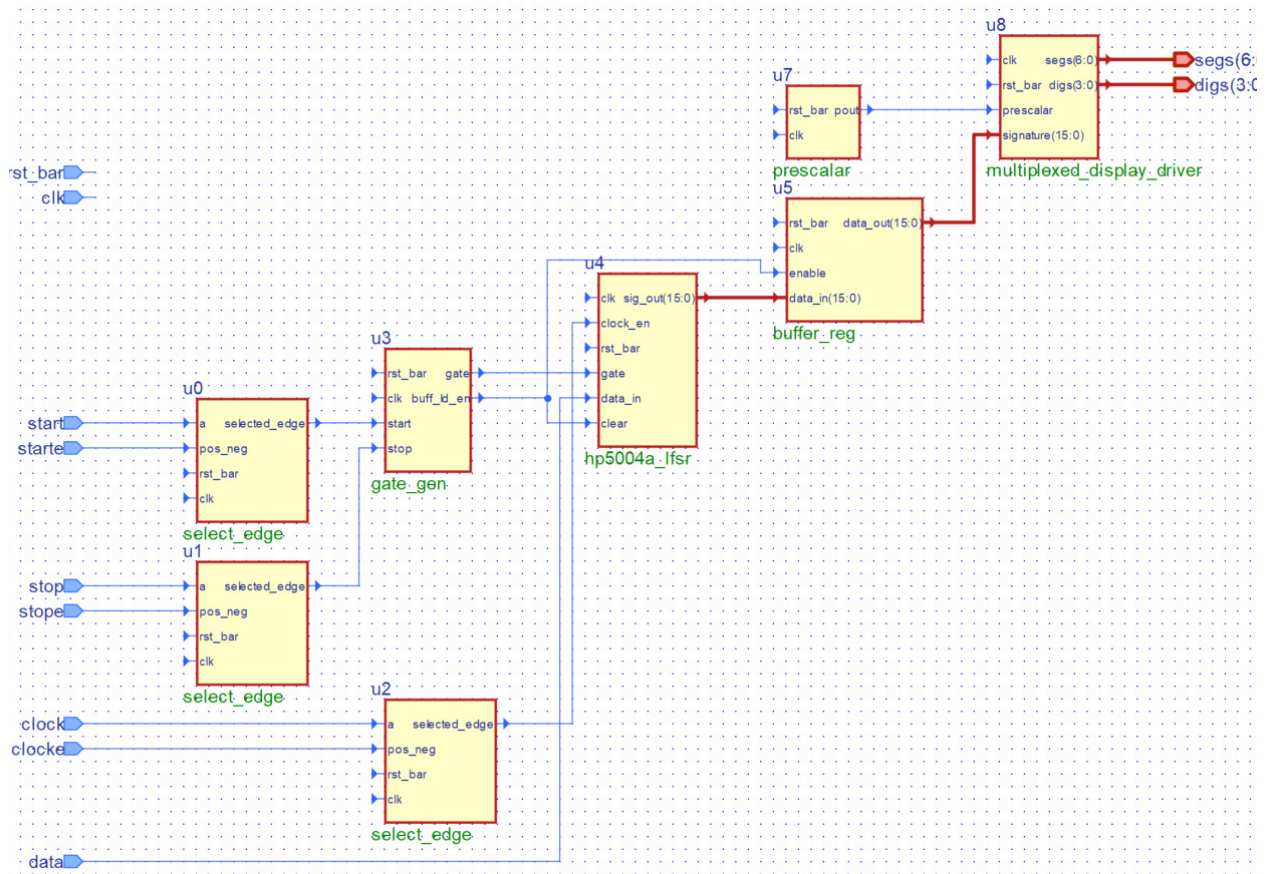
**Submit your properly commented design description and testbench source files as part of your prelab.**

#### ***Design Task 4: Structural Signature Analyzer with Added Features***

Modify your `hp5004a_SA_w_display` design from Task 2 of Laboratory 11 to include the additional features previously described in this laboratory document.

In Laboratory 11 Task 2, your system received a single gate for each test. In normal operation, the HP5004A is continually receiving start and stop signals in essentially a loop. This means that gates are continually generated during a measurement. If you connect the `sig_out` output from `hp5004A_lfsr` directly to the `signature` input of `multiplexed_display_driver`, the displayed hex digits will be constantly changing, since gates will continually be generated. You would not be able to read the result on the 4-digit seven segment display.

To prevent this you must add a buffer register between `sig_out` from `hp5004A_lfsr` and the `signature` input of the `multiplexed_display_driver`. This buffer register only stores `sig_out` at the end of the gate signal (falling edge). Name this entity `buffer_reg`.



Write a testbench that can generate simulated signals from the DUT to your signature analyzer.

**Submit your properly commented complete design description and testbench source files as part of your prelab.**